

MH3100 Real Analysis I

Midterm Practice 4 (from past Finals) (Solutions)

Past Final questions in Midterm scope (Lectures 1–17)

Quantitative Research Society @NTU

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About This Practice Paper

This is QRS@NTU's fourth MH3100 Midterm practice paper. Instead of a fresh QRS-authored paper, it collects every question part from NTU's three most recent MH3100 Real Analysis I *Final* examinations (AY2022/23, AY2023/24, AY2024/25, Semester 2) that falls inside the **Midterm scope, Lectures 1–17** (construction and completeness of $\mathbb{R}$ through sets of real numbers). Parts that require Lecture 18–20 material (function limits, continuity, uniform continuity), differentiation, integration, uniform convergence of function sequences/series, or open covers/Heine-Borel are outside the Midterm syllabus and are omitted below; the scope table gives the reason for every omission. Since this paper draws only on genuine past-Final material, its total (97 marks across 8 questions) does not match the 24-mark, 60-minute Midterm format – treat it as extra practice, not a timed mock.

Solutions in this paper are held to the same three writing expectations as the Midterm: complete sentences; related statements joined by logical connectives; and inference explicitly marked with "therefore/hence/so".

Overview

Papers used (all MH3100 Real Analysis I, NTU, CLOSED BOOK, 6 questions, 2 hours):

- Semester 2 Examination 2022–2023 (April 2023).
- Semester II Examination 2023–2024 (April 2024).
- Semester II Examination 2024–2025 (April 2025).

In every one of the three papers, Question 1 (15 marks, five True/False parts) and Question 2 (15 marks, three existential parts) mix in-scope and out-of-scope sub-parts, Question 3 (20 marks) is a supremum/infimum proof that is fully in scope, and Questions 4–6 (20+20+10 marks) all hinge on continuity and/or uniform convergence and are fully out of scope. Marks are not printed per sub-part on the original papers for Questions 1–2; QRS distributes each question's total evenly across its parts.

Scope table

AY2022/23 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Monotone Convergence Lemma (L9); Theorem on Subsequential Limits (L7); limit points (L13) – all in scope.
Q1(d)–(e)	Out	Uniform continuity; power series and uniform convergence – beyond L17.
Q2(a)	Partly (1/3)	Absolute convergence + bounded sequence, via Comparison/Absolute Convergence Test (L12).
Q2(b)–(c)	Out	Function limits (L18–20 topic); pointwise convergence of continuous functions – beyond L17.
Q3	In	Definition of supremum (L2), fully in scope.
Q4	Out	Hypothesis “ f continuous” is L18–20 material.
Q5	Out	Uniform convergence of a function series – not covered in L1–17.
Q6	Out	Uniform convergence of continuous functions – not covered in L1–17.

AY2023/24 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Suprema of sets (L2); Bolzano-Weierstrass Theorem (L8); isolated points/closed sets (L13, L15) – all in scope.
Q1(d)–(e)	Out	Differentiability; uniform convergence of derivatives – beyond L17.
Q2(a)	Partly (1/3)	Series with $\sum a_n$ convergent but $\sum a_n^2$ not (L10–12).
Q2(b)–(c)	Out	Continuity and uniform continuity – beyond L17.
Q3	In	Definition of supremum (L2), fully in scope.
Q4	Out	Continuous functions f, g – L18–20 material.
Q5	Out	Uniform convergence of a function series.
Q6	Out	Function limits (L18–20 topic).

AY2024/25 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Sup/inf (L2); Cauchy Criterion for Series (L10) with the Archimedean Property (L3); closure/closed sets (L14–15) – all in scope.
Q1(d)–(e)	Out	Function limits (L18–20 topic); Weierstrass- M -test-style uniform convergence.
Q2	Out (0/3)	(a)–(b) continuous-function hypotheses; (c) uniform convergence – all beyond L17.
Q3	In	Sup/inf via rational upper bounds (L2) + Archimedean Property (L3), fully in scope.
Q4	Out	Continuous functions and compactness combined; the continuity hypothesis is L18–20 material (compactness itself, L16, would be in scope).
Q5	Out	Uniform convergence of a function series.
Q6	Out	Continuity + uniform convergence.

Total marks in this practice paper

Question	1	2	3	4	5	6	7	8	Total
Marks	9	5	20	9	5	20	9	20	97

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Question 1

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) A monotone sequence is convergent if and only if it has a convergent subsequence.
- (b) A sequence is divergent if it has a divergent subsequence.
- (c) If a set has no limit points, then it is finite.

(Source: MH3100 Final AY22-23 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on uniformly continuous functions and uniform convergence of power series, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. The Final prints only the question total, 15 marks, for all five parts; QRS distributes this evenly, 3 marks per part.)

Solution

- (a) ☒ True

Justification. We prove both directions. If $\{a_n\}$ is convergent, then $\{a_n\}$ is itself a subsequence of itself (take $n_k = k$ for each $k \in \mathbb{N}$), so $\{a_n\}$ has a convergent subsequence. Conversely, suppose $\{a_n\}$ is monotone and has a convergent subsequence. The Monotone Convergence Lemma states that an eventually monotone sequence with a convergent subsequence is convergent; since $\{a_n\}$ is monotone and has a convergent subsequence by hypothesis, the Monotone Convergence Lemma implies that $\{a_n\}$ is convergent. Therefore $\{a_n\}$ is convergent if and only if it has a convergent subsequence, so the statement is true. $\square$

- (b) ☒ True

Justification. Suppose $\{a_n\}$ has a divergent subsequence $\{a_{n_k}\}$. Assume, for the sake of contradiction, that $\{a_n\}$ is convergent, with limit L . The Theorem on Subsequential Limits states that every subsequence of a convergent sequence converges to the same limit as the sequence itself; therefore $\{a_{n_k}\}$, being a subsequence of $\{a_n\}$, would converge to L . This contradicts the hypothesis that $\{a_{n_k}\}$ is divergent. Hence $\{a_n\}$ is not convergent, that is, $\{a_n\}$ is divergent, so the statement is true. $\square$

- (c) ☐ False

Justification. (Scratch work – not part of the written answer.) Take $F = \mathbb{Z}$: consecutive integers are always exactly 1 apart, so no real number can have two integers within $1/2$ of it, which is enough to rule out every candidate limit point.

We give a counterexample: let $F := \mathbb{Z}$, which is infinite. We first show F has no limit points. Let $a \in \mathbb{R}$ be arbitrary. Since any two distinct integers differ by at least 1 in absolute value, at most one integer can lie in the interval $(a - \frac{1}{2}, a + \frac{1}{2})$. If no integer lies in this interval, then, taking $\varepsilon = \frac{1}{2}$, there is no $x \in F \setminus \{a\}$ with $|x - a| < \varepsilon$, so a is not a limit point of F . If exactly one integer x_0 lies in this interval, we consider two cases. If $x_0 = a$, then x_0 is excluded from $F \setminus \{a\}$, so $\varepsilon = \frac{1}{2}$ again shows a is not a limit point of F . If $x_0 \neq a$, then, taking $\varepsilon := |x_0 - a| > 0$, the point x_0 satisfies $|x_0 - a| \not< \varepsilon$, and since x_0 was the unique integer within $\frac{1}{2}$ of a and $\varepsilon \leq \frac{1}{2}$, no other integer lies within ε of a .

a either; hence again there is no $x \in F \setminus \{a\}$ with $|x - a| < \varepsilon$. In every case, some $\varepsilon > 0$ witnesses that a is not a limit point of F . Since $a \in \mathbb{R}$ was arbitrary, $F = \mathbb{Z}$ has no limit points. However, F is infinite. Therefore a set having no limit points need not be finite, so the statement is false. $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False. No justification is required on the actual exam.
- (b) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.
- (c) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.

Typical deductions (if justification is examined): in (a), proving only one direction of the "if and only if"; in (b), attempting to prove convergence directly instead of arguing by contradiction via the Theorem on Subsequential Limits; in (c), searching only among finite sets instead of recognising that infinite discrete sets such as $\mathbb{Z}$ or $\mathbb{N}$ already settle the claim.

Question 2

(5 Marks)

For each of the following existential statements, either give an example fulfilling the statement or prove that it is not true.

There is an absolutely convergent series $\sum_{n=1}^{\infty} a_n$ and bounded sequence $\{b_n\}_{n=1}^{\infty}$ for which the series $\sum_{n=1}^{\infty} a_n b_n$ is divergent.

(Source: MH3100 Final AY22-23 Sem 2, Question 2(a). Question 2(b)–(c), on function limits and pointwise convergence of continuous functions, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. The Final's Question 2 total, 15 marks, is distributed evenly, 5 marks per part.)

Solution

Idea. (Scratch work – not part of the written answer.) If $|b_n| \leq M$ for every n , then $|a_n b_n| \leq M|a_n|$, so $\sum |a_n b_n|$ is dominated term-by-term by the convergent series $\sum M|a_n|$; the Comparison Test then forces $\sum a_n b_n$ to converge absolutely, so no divergent example can exist.

Proof. We show the statement is not true: for every absolutely convergent series $\sum_{n=1}^{\infty} a_n$ and every bounded sequence $\{b_n\}$, the series $\sum a_n b_n$ converges.

Let $\sum_{n=1}^{\infty} a_n$ be absolutely convergent, that is, $\sum |a_n|$ is convergent, and let $\{b_n\}$ be a bounded sequence. By the definition of a bounded sequence, there exists $M > 0$ such that $|b_n| \leq M$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}$,

$$0 \leq |a_n b_n| = |a_n| |b_n| \leq M |a_n|.$$

Since $\sum |a_n|$ is convergent, the Algebraic Limit Theorem for Series (scalar multiple) implies that $\sum M|a_n|$ is also convergent. Since $0 \leq |a_n b_n| \leq M|a_n|$ for every $n \in \mathbb{N}$ and $\sum M|a_n|$ is convergent, the Comparison Test implies that $\sum |a_n b_n|$ is convergent, that is, $\sum_{n=1}^{\infty} a_n b_n$ is absolutely convergent. Therefore, by the Absolute Convergence Test, $\sum_{n=1}^{\infty} a_n b_n$ is convergent.

Hence, for every absolutely convergent series $\sum_{n=1}^{\infty} a_n$ and every bounded sequence $\{b_n\}$, the series $\sum a_n b_n$ converges. So no absolutely convergent series and bounded sequence can produce a divergent $\sum a_n b_n$, and the statement is false. $\square$

Common slip. Attempting to build a counterexample using a conditionally (but not absolutely) convergent series such as $a_n = (-1)^n / \sqrt{n}$, which is not eligible here; forgetting that boundedness of $\{b_n\}$, not merely its convergence, is exactly what the Comparison Test needs.

Marking scheme (5 marks)

QRS suggested marking scheme.

- **1 mark:** correctly recognising the statement is false, not attempting to fabricate an example.
- **1 mark:** introducing the bound $M > 0$ with $|b_n| \leq M$ and the pointwise estimate $|a_n b_n| \leq M|a_n|$.
- **1 mark:** correct use of the Comparison Test to conclude $\sum |a_n b_n|$ converges.
- **1 mark:** correct use of the Absolute Convergence Test to conclude $\sum a_n b_n$ converges.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: citing the Comparison Test with the inequality reversed; concluding convergence of $\sum a_n b_n$ without the intermediate absolute-convergence step; using a non-bounded or only conditionally convergent example.

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Question 3

(20 Marks)

Use the definition of supremum to prove that if two nonempty sets $A, B \subseteq \mathbb{R}$ that are bounded above satisfy $\sup A < \sup B$, then for all $a \in A$, there exists $b \in B$ such that $a < b$. Give an example to show that the converse is not true.

You do not need to show that the suprema exist.

(Source: MH3100 Final AY22-23 Sem 2, Question 3, verbatim, 20 marks.)

Solution

Method 1: Direct argument from the two defining properties of supremum

Idea. (*Scratch work – not part of the written answer.*) Let $s = \sup A < t = \sup B$ and let $a \in A$. If a were an upper bound for B , then $t \leq a$ by minimality of t among upper bounds of B ; but $a \leq s < t$, giving $t \leq a < t$, a contradiction. So a is not an upper bound for B , which is exactly what furnishes $b \in B$ with $b > a$. For the converse, look for two sets with the *same* supremum, neither attained, so that "for all a there is a bigger b " still holds without $\sup A < \sup B$.

Proof. Let $s := \sup A$ and $t := \sup B$, and suppose $s < t$. Let $a \in A$ be arbitrary. Since $s = \sup A$ is an upper bound for A , $a \leq s$.

Suppose, for the sake of contradiction, that there is no $b \in B$ with $a < b$; that is, every $b \in B$ satisfies $b \leq a$. Then a is an upper bound for B . Since $t = \sup B$ is not greater than any upper bound for B , this implies $t \leq a$. Combining $t \leq a$ with $a \leq s < t$ gives $t \leq a \leq s < t$, that is, $t < t$, which is false.

Therefore the assumption is false, so there exists $b \in B$ with $a < b$. Since $a \in A$ was arbitrary, for every $a \in A$ there exists $b \in B$ with $a < b$. $\square$

Example (converse is not true). Let $A := (0, 1)$ and $B := (0, 1)$. For every $a \in A$, the number $b := \frac{a+1}{2}$ satisfies $a < b < 1$, so $b \in B$ and $a < b$; hence for every $a \in A$ there exists $b \in B$ with $a < b$. However, $\sup A = 1 = \sup B$, so $\sup A < \sup B$ is false. This shows the converse of the statement – "if for all $a \in A$ there exists $b \in B$ with $a < b$, then $\sup A < \sup B$ " – is not true.

Common slip. Concluding $t \leq a$ directly from " a is an upper bound for B " without citing that $t = \sup B$ is the *least* such upper bound; producing a converse "counterexample" in which $\sup A < \sup B$ still happens to hold (e.g. using disjoint intervals), which fails to contradict the converse.

Marking scheme (20 marks)

QRS suggested marking scheme.

- **4 marks:** correct definition of supremum (upper bound; not greater than any upper bound) applied to both $s = \sup A$ and $t = \sup B$.
- **4 marks:** setting up the proof by contradiction, assuming a is an upper bound for B .
- **4 marks:** correctly deriving $t \leq a$ from the minimality of $t = \sup B$ among upper bounds of B .
- **4 marks:** correctly combining $t \leq a \leq s < t$ to reach the contradiction, and concluding the existence of $b \in B$ with $a < b$.

- **3 marks:** a valid example (with justification) showing the converse fails.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: proving only that a is not an upper bound for B without extracting the required $b \in B$ with $a < b$; choosing a converse example in which $\sup A < \sup B$ still holds.

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Question 4

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) If the set A has supremum s and the set B has supremum t , then their intersection $A \cap B$ has supremum $\min\{s, t\}$.
- (b) If an unbounded sequence has a bounded subsequence, then it has a convergent subsequence.
- (c) If a set contains only isolated points, then it is closed.

(Source: MH3100 Final AY23-24 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on differentiability and uniform convergence of derivatives, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks are again distributed evenly, 3 marks per part.)

Solution

- (a) False

Justification. Let $A := (0, 1)$ and $B := (2, 3)$. Then $s := \sup A = 1$ and $t := \sup B = 3$, so $\min\{s, t\} = 1$. However $A \cap B = \emptyset$, and, as noted after the Axiom of Completeness, the empty set does not have a real supremum, since every real number is vacuously an upper bound for $\emptyset$ and there is no least one. Since $A \cap B$ does not have supremum $1 = \min\{s, t\}$ (indeed it has no real supremum at all), the statement is false. $\square$

- (b) True

Justification. Let $\{a_n\}$ be an unbounded sequence with a bounded subsequence $\{a_{n_k}\}$. Since $\{a_{n_k}\}$ is bounded, the Bolzano-Weierstrass Theorem implies that $\{a_{n_k}\}$ has a convergent subsequence $\{a_{n_{k_j}}\}$. Since a subsequence of a subsequence of $\{a_n\}$ is itself a subsequence of $\{a_n\}$, the sequence $\{a_{n_{k_j}}\}$ is a convergent subsequence of $\{a_n\}$. Hence $\{a_n\}$ has a convergent subsequence, so the statement is true. $\square$

- (c) False

Justification. (Scratch work – not part of the written answer.) Take $F = \{1/n : n \in \mathbb{N}\}$: every point of F sits at a definite gap from its neighbours in F (so is isolated), yet 0 is a limit point of F that is not in F .

Let $F := \{1/n : n \in \mathbb{N}\}$. We first show every point of F is an isolated point of F . Let $n \in \mathbb{N}$ and $a := 1/n \in F$, and take $\varepsilon := \frac{1}{n(n+1)}$. For every $m \in \mathbb{N}$ with $m \neq n$: if $m > n$ (so $m \geq n+1$), then $\frac{1}{m} \leq \frac{1}{n+1}$, so $a - \frac{1}{m} \geq \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} = \varepsilon$; if $m < n$ (so $n \geq 2$ and $m \leq n-1$), then $\frac{1}{m} \geq \frac{1}{n-1}$, so $\frac{1}{m} - a \geq \frac{1}{n-1} - \frac{1}{n} = \frac{1}{n(n-1)} \geq \frac{1}{n(n+1)} = \varepsilon$. In either case $|\frac{1}{m} - a| \geq \varepsilon$, so no $x \in F \setminus \{a\}$ satisfies $|x - a| < \varepsilon$; hence a is not a limit point of F , that is, a is an isolated point of F . Since $n \in \mathbb{N}$ was arbitrary, F contains only isolated points.

However, 0 is a limit point of F : given $\varepsilon > 0$, the Archimedean Property gives $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$, so $\frac{1}{N} \in F \setminus \{0\}$ satisfies $|\frac{1}{N} - 0| < \varepsilon$; since $\varepsilon > 0$ was arbitrary, 0 is a limit point of F . Since $0 \notin F$, F does not contain all its limit points, so F is not closed. Hence a set containing only isolated points need not be closed, so the statement is false. $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.
- (b) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.
- (c) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.

Typical deductions: in (a), forgetting that an empty intersection has no real supremum; in (b), forgetting that a subsequence of a subsequence is still a subsequence of the original sequence; in (c), confusing "every point is isolated within the set" with "the set has no limit points at all outside itself".

Question 5

(5 Marks)

For each of the following existential statements, either give an example fulfilling the statement or prove that it is not true.

There is a convergent series $\sum_{n=1}^{\infty} a_n$ such that the series $\sum_{n=1}^{\infty} a_n^2$ is not convergent.

(Source: MH3100 Final AY23-24 Sem 2, Question 2(a). Question 2(b)–(c), on continuous functions and Cauchy sequences and on uniform continuity, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks distributed evenly, 5 marks per part.)

Solution

Idea. (Scratch work – not part of the written answer.) Squaring kills the alternation that made an only-conditionally-convergent series converge in the first place: take $a_n := (-1)^n/\sqrt{n}$, which decays to 0 but is not absolutely summable, so $\sum a_n$ converges by the Alternating Series Test while $\sum a_n^2 = \sum 1/n$ is the divergent harmonic series.

Construction. Let $a_n := \frac{(-1)^n}{\sqrt{n}}$ for each $n \in \mathbb{N}$.

Proof. We first show $\sum a_n$ is convergent. The sequence $\left\{\frac{1}{\sqrt{n}}\right\}$ is decreasing, since $n \mapsto \sqrt{n}$ is increasing, and converges to 0: given $\varepsilon > 0$, the Archimedean Property gives $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon^2$, so for all $n \geq N$, $\frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon$. Therefore, by the Alternating Series Test, $\sum a_n = \sum (-1)^n/\sqrt{n}$ is convergent.

We next show $\sum a_n^2$ is divergent. For every $n \in \mathbb{N}$, $a_n^2 = 1/n$, so $\sum a_n^2$ is the harmonic series $\sum 1/n$. For each $m \in \mathbb{N}$, the partial sums of the harmonic series satisfy

$$s_{2m} - s_m = \frac{1}{m+1} + \frac{1}{m+2} + \cdots + \frac{1}{2m} \geq m \cdot \frac{1}{2m} = \frac{1}{2},$$

since each of the m terms in the sum is at least $\frac{1}{2m}$. Taking $\varepsilon = \frac{1}{2}$, this shows no $N \in \mathbb{N}$ can make the tail sum from $N+1$ to $2N$ smaller than ε , so the sequence of partial sums of $\sum 1/n$ fails the Cauchy Criterion for Series; hence $\sum a_n^2 = \sum 1/n$ is divergent.

Therefore $\sum a_n$ is a convergent series for which $\sum a_n^2$ is not convergent, so the statement is true. $\square$

Common slip. Choosing an absolutely convergent example, such as $a_n = 1/n^2$, for which $a_n^2 = 1/n^4$ also converges, so the example fails to witness the statement; misapplying the Alternating Series Test to a sequence that is not decreasing.

Marking scheme (5 marks)

QRS suggested marking scheme.

- **1 mark:** a valid example built on conditional-but-not-absolute convergence.
- **2 marks:** correct proof that $\sum a_n$ converges (Alternating Series Test, with the hypothesis that the terms decrease to 0 verified).
- **1 mark:** correct proof that $\sum a_n^2 = \sum 1/n$ diverges (Cauchy Criterion for Series, tail estimate).
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: an example where both series behave the same way; omitting the decreasing-terms check before invoking the Alternating Series Test.

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Question 6

(20 Marks)

Use the definition of supremum to prove that if a set $A \subseteq \mathbb{R}$ is nonempty and bounded above and B denotes the set $\{b \in \mathbb{R} : \exists a \in A, b < a\}$, then B has a supremum $\sup B$ and the supremum satisfies $\sup B = \sup A$.

(Source: MH3100 Final AY23-24 Sem 2, Question 3, verbatim, 20 marks. Unlike Question 3 of the other two Finals in this paper, this question does not waive the requirement to show the supremum exists.)

Solution

Method 1: Establish existence, then squeeze $\sup B$ between $\sup A$ both ways

Idea. (Scratch work – not part of the written answer.) B is the set of all reals lying strictly below some element of A , i.e. $B = \bigcup_{a \in A} (-\infty, a)$. Since A is bounded above, the same bound bounds B above, so the Axiom of Completeness gives $\sup B$. To show $\sup B = \sup A =: s$: s is an upper bound of B (easy, using $b < a \leq s$), and no smaller number can be an upper bound of B , else it would fail to be an upper bound of A and so be beaten by some $a \in A$, which manufactures an even larger element of B .

Proof. Write $s := \sup A$, which exists by the Axiom of Completeness since A is nonempty and bounded above.

Step 1: B is nonempty. Since A is nonempty, there is $a_0 \in A$. The number $b_0 := a_0 - 1$ satisfies $b_0 < a_0$ with $a_0 \in A$, so $b_0 \in B$. Hence $B \neq \emptyset$.

Step 2: B is bounded above by s . Let $b \in B$ be arbitrary. By the definition of B , there exists $a \in A$ with $b < a$. Since $s = \sup A$ is an upper bound for A , $a \leq s$. Therefore $b < a \leq s$, so $b \leq s$. Since $b \in B$ was arbitrary, s is an upper bound for B .

By Steps 1 and 2, B is a nonempty set bounded above; therefore the Axiom of Completeness implies B has a supremum. Write $t := \sup B$.

Step 3: $t \leq s$. Since s is an upper bound for B (Step 2) and $t = \sup B$ is not greater than any upper bound for B , we have $t \leq s$.

Step 4: $s \leq t$. Suppose, for the sake of contradiction, that $s > t$. Since $t = \sup B$ is an upper bound for B , every $b \in B$ satisfies $b \leq t$. Since $t < s = \sup A$, the number t is not an upper bound for A : if it were, then, since $s = \sup A$ is not greater than any upper bound for A , we would have $s \leq t$, contradicting $t < s$. Since t is not an upper bound for A , there exists $a \in A$ with $a > t$. Let $b^* := \frac{t+a}{2}$; then $t < b^* < a$, so $b^* < a$ with $a \in A$, giving $b^* \in B$. But $b^* > t$, contradicting that every element of B satisfies $b \leq t$.

This contradiction shows $s \leq t$.

By Steps 3 and 4, $t = s$, that is, $\sup B = \sup A$. □

Common slip. Invoking the Axiom of Completeness for $\sup B$ before establishing that B is nonempty and bounded above; in Step 4, picking $b^* = t$ or $b^* = a$ instead of a point strictly between them, either of which fails to witness the needed strict inequality.

Marking scheme (20 marks)

QRS suggested marking scheme.

- **3 marks:** Step 1 (B nonempty).
- **4 marks:** Step 2 (B bounded above by $s = \sup A$), with the definition of B correctly unpacked.

- **3 marks:** correctly invoking the Axiom of Completeness to obtain $t = \sup B$.
- **4 marks:** Step 3 ($t \leq s$).
- **5 marks:** Step 4 ($s \leq t$), including the proof-by-contradiction structure and the midpoint construction b^* .
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: asserting $\sup B$ exists without first showing B is nonempty and bounded above; omitting the contrapositive step " $t < s \implies t$ is not an upper bound for A " in Step 4.

Question 7

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) Two sets $A, B \subseteq \mathbb{R}$ that satisfy $(\forall a \in A, \forall b \in B, a < b)$ must satisfy $\sup A < \inf B$.
- (b) A series $\sum_{n=1}^{\infty} a_n$ that satisfies $(m > n \implies |a_{n+1} + a_{n+2} + \cdots + a_m| < \frac{1}{n})$ for all natural numbers $m, n \in \mathbb{N}$ is convergent.
- (c) A set is closed if and only if it is the closure of some set.

(Source: MH3100 Final AY24-25 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on function limits and uniform convergence of a series of functions, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks distributed evenly, 3 marks per part.)

Solution

- (a) False

Justification. Let $A := (0, 1)$ and $B := (1, 2)$. For every $a \in A$ and every $b \in B$, $a < 1 < b$, so $a < b$; hence A, B satisfy the hypothesis. However, $\sup A = 1$ and $\inf B = 1$, so $\sup A < \inf B$ reads $1 < 1$, which is false. Hence the statement is false. $\square$

- (b) True

Justification. (Scratch work – not part of the written answer.) The hypothesis is a strengthened Cauchy Criterion for Series in which the bound $1/n$, rather than an arbitrary ε , controls the tail sums; since $1/n \rightarrow 0$, any $\varepsilon > 0$ is eventually beaten by $1/n$, recovering the ordinary Cauchy Criterion.

Let $\varepsilon > 0$ be given. By the Archimedean Property, there exists $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$. Let $n, m \in \mathbb{N}$ satisfy $m > n \geq N$. By hypothesis, applied to this n and m , and using $n \geq N$ for the middle inequality,

$$|a_{n+1} + a_{n+2} + \cdots + a_m| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this is exactly the Cauchy Criterion for Series, so $\sum a_n$ is convergent. Hence the statement is true. $\square$

- (c) True

Justification. (Scratch work – not part of the written answer.) $(\Leftarrow)$ A closure $\overline{A}$ is always closed, since $\overline{\overline{A}} = \overline{A}$ (the Corollary on Closures of Closures) already puts it into the form the Closure Criterion for Closed Sets requires. $(\Rightarrow)$ If F is closed then $\overline{F} = F$ by the same Closure Criterion, so F is the closure of the set F itself.

$(\Leftarrow)$ Suppose $F = \overline{A}$ for some set $A \subseteq \mathbb{R}$. By the Corollary on Closures of Closures, $\overline{\overline{A}} = \overline{A}$, that is, $\overline{F} = F$. Since F contains its own closure, the Closure Criterion for Closed Sets implies F is closed.

$(\Rightarrow)$ Suppose F is closed. By the Closure Criterion for Closed Sets, F is closed if and only if $F \supseteq \overline{F}$; since F is closed, $\overline{F} \subseteq F$. On the other hand, the closure of any set always

contains that set, so $F \subseteq \overline{F}$. Combining both inclusions, $\overline{F} = F$, so F is the closure of the set F itself.

Hence a set is closed if and only if it is the closure of some set, so the statement is true.
 $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.
- (b) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.
- (c) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.

Typical deductions: in (a), confusing strict inequality between every pair with strict inequality between the sup and inf; in (b), forgetting to choose N via the Archimedean Property so that $1/N < \varepsilon$; in (c), proving only one direction of the biconditional.

Question 8

(20 Marks)

Use the definitions of suprema and infima to prove that if a set $A \subseteq \mathbb{R}$ and its set $B := \{b \in \mathbb{Q} : \forall a \in A, a \leq b\}$ of all rational upper bounds are nonempty, then $\sup A = \inf B$.

You do not need to show that the supremum and infimum exist.

(Source: MH3100 Final AY24-25 Sem 2, Question 3, verbatim, 20 marks.)

Solution

Method 1: Squeeze $\sup A$ and $\inf B$ together, using an explicit rational between two reals

Idea. (*Scratch work – not part of the written answer.*) Write $s := \sup A$, $t := \inf B$. Every $b \in B$ is, in particular, a real upper bound for A , so $s \leq b$ for every $b \in B$; this makes s a lower bound for B , giving $s \leq t$. For the reverse inequality, suppose $t > s$; a rational q with $s \leq q < t$ would be a rational upper bound for A (since $q \geq s \geq a$ for every $a \in A$), so $q \in B$, yet $q < t = \inf B$ would contradict t being a lower bound for B . Producing such a q needs a rational strictly between two given reals, built explicitly from the Archimedean Property since no density-of- $\mathbb{Q}$ theorem is available to cite.

Lemma (a rational between two reals). If $x, y \in \mathbb{R}$ and $x < y$, then there is $q \in \mathbb{Q}$ with $x < q < y$.

Proof of Lemma. By the Archimedean Property, there is $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$. By the Archimedean Property again, there is $M \in \mathbb{N}$ with $M > |nx|$, so $nx + M > 0$. Let $S := \{k \in \mathbb{N} : k > nx + M\}$. By the Archimedean Property, S is nonempty. Since S is a nonempty subset of $\mathbb{N}$, it has a least element k_0 . Then $k_0 > nx + M$; and, since k_0 is least in S , either $k_0 = 1$ (so $nx + M < 1 \leq k_0$) or $k_0 - 1 \notin S$, which means $k_0 - 1 \leq nx + M$. In either case $k_0 - 1 \leq nx + M < k_0$, since $nx + M > 0$ covers the case $k_0 = 1$ too.

Let $k := k_0 - 1 - M \in \mathbb{Z}$. From $k_0 - 1 \leq nx + M$ we get $k \leq nx$; from $nx + M < k_0$ we get $nx < k_0 - M = k + 1$. Hence $k \leq nx < k + 1$.

Let $q := \frac{k+1}{n} \in \mathbb{Q}$. From $nx < k + 1$, dividing by $n > 0$ gives $x < q$. From $k \leq nx$, dividing by n gives $q = \frac{k}{n} + \frac{1}{n} \leq x + \frac{1}{n}$; since $\frac{1}{n} < y - x$, this gives $q \leq x + \frac{1}{n} < x + (y - x) = y$. Hence $x < q < y$. $\square$

Proof (of the main statement). Write $s := \sup A$ and $t := \inf B$.

Step 1: $s \leq t$. Let $b \in B$ be arbitrary. By the definition of B , $b \in \mathbb{Q} \subseteq \mathbb{R}$ and $a \leq b$ for every $a \in A$, so b is a (real) upper bound for A . Since $s = \sup A$ is not greater than any upper bound for A , $s \leq b$. Since $b \in B$ was arbitrary, s is a lower bound for B . Since $t = \inf B$ is not less than any lower bound for B , $s \leq t$.

Step 2: $t \leq s$. Suppose, for the sake of contradiction, that $t > s$. By the Lemma, applied with $x = s$ and $y = t$, there is $q \in \mathbb{Q}$ with $s < q < t$. For every $a \in A$, since $s = \sup A$ is an upper bound for A , $a \leq s < q$, so $a \leq q$. Since $q \in \mathbb{Q}$ and q is an upper bound for A , the definition of B gives $q \in B$. Since $t = \inf B$ is a lower bound for B , $t \leq q$. But $q < t$, so $t \leq q < t$, that is, $t < t$, which is false.

Therefore the assumption is false, so $t \leq s$.

By Steps 1 and 2, $s = t$, that is, $\sup A = \inf B$. $\square$

Common slip. Forgetting that B consists of *rational* upper bounds and constructing an arbitrary real q instead of a rational one; treating "there is a rational between two reals" as obvious without proof, despite no such theorem appearing in Lectures 1–17 (it must be built from the Archimedean Property, as in the Lemma).

Marking scheme (20 marks)

QRS suggested marking scheme.

- **3 marks:** Step 1 ($s \leq t$) complete.
- **3 marks:** Lemma correctly set up (choosing n with $1/n < y-x$, and M with $nx+M > 0$).
- **6 marks:** Lemma's construction of k via the least element of S , and the verification $k \leq nx < k+1$.
- **3 marks:** correctly applying the Lemma in Step 2 (the rational q satisfies $s < q < t$, is an upper bound for A , and hence $q \in B$).
- **4 marks:** deriving the contradiction $t \leq q < t$ and concluding $t \leq s$.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: skipping the existence argument for the least element k_0 of S ; using a real (not necessarily rational) q in Step 2, which does not lie in B .